

Indian Iron & Steel Sector Skill Council

QUALIFICATIONS PACK - OCCUPATIONAL STANDARDS FOR IRON & STEEL INDUSTRY

What are Occupational Standards (OS)?

- OS describe what individuals need to do, know and understand in order to carry out a particular job role or function
- OS are performance standards that individuals must achieve when carrying out functions in the workplace, together with specifications of the underpinning knowledge and understanding



Contents

1. Introduction and Contacts.....P.1
2. Qualifications Pack.....P.2
3. Glossary of Key TermsP.4
4. NOS Units.....P.6

Introduction

Qualifications Pack- Process Operator Rolling Mills

SECTOR: Iron & Steel

SUB-SECTOR: Steel, Re-Rollers

REFERENCE ID: ISC/Q0701

ALIGNED TO: NCO-2004/NIL

Title of Job: This job is all about keeping different process lines functional by ensuring continuity of process. It also involves providing the actual commands for different operations as needed and monitor the operations for any deviations. It includes inspecting the material between processes to ensure alignment to specifications.

Personal Attributes: This job requires the candidate to work independently as well as in teams. Some of the key attributes includes physical fitness, not having colour blindness, having analytical skills, problem solving attitude, high concentration levels, a sharp reflex and willingness to work in a factory environment.



Job Details

Qualifications Pack Code	ISC/Q0701		
Job Role	Process Operator: Rolling Mills		
Credits(NSQF)	TBD	Version number	1.0
Industry	Iron & Steel	Drafted on	18/09/2014
Sub-sector	Steel, Re-Rollers	Last reviewed on	30/12/2014
Occupation	Cold Rolling Mill	Next review date	30/12/2015
NSQC Clearance on	18/06/2015		

Job Role	Process Operator: Rolling Mills
Role Description	This involves keeping the different process lines (viz. Pickling, Rolling, Annealing, Galvanising, Utilities etc.) functional by ensuring continuity of process. This includes providing the actual commands for different operations, as needed and monitors the operations for any deviations. This also involves inspection of the material between the workstations of the processes to ensure alignment to specifications.
NSQF level	5
Minimum Educational Qualifications	ITI pass
Maximum Educational Qualifications	Diploma pass
Training (Suggested but not mandatory)	- Basics of metallurgy - Acid handling - Annealing - Galvanising - Basics of Hydraulics, Pneumatics and Electrical controls
Minimum Job Entry Age	18 years
Experience	0-1 year experience in similar function - In lieu of minimum qualification the incumbent should have 4-5 years of relevant work experience under an experienced Process Operator



Occupational Standards (OS)	Compulsory: ISC/N0706: Prepare for operation of control systems ISC/N0707: Perform operation of control systems ISC/N0008: Use basic health and safety practices at the workplace ISC/N0096: Works effectively with others and manage team members Optional: N/A
Performance Criteria	As described in the relevant NOS units



Definitions

Keywords /Terms	Description
Core Skills/Generic Skills	Core Skills or Generic Skills are a group of skills that are key to learning and working in today's world. These skills are typically needed in any work environment. In the context of the NOS, these include communication related skills that are applicable to most job roles.
Function	Function is an activity necessary for achieving the key purpose of the sector, occupation, or area of work, which can be carried out by a person or a group of persons. Functions are identified through functional analysis and form the basis of NOS.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organization.
Knowledge and Understanding	Knowledge and Understanding are statements which together specify the technical, generic, professional and organizational specific knowledge that an individual needs in order to perform to the required standard.
National Occupational Standards (NOS)	NOS are Occupational Standards which apply uniquely in the Indian context
Occupation	Occupation is a set of job roles, which perform similar/related set of functions in an industry.
Organisational Context	Organisational Context includes the way the organization is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Performance Criteria	Performance Criteria are statements that together specify the standard of performance required when carrying out a task.
Qualifications Pack(QP)	Qualifications Pack comprises the set of NOS, together with the educational, training and other criteria required to perform a job role. A Qualifications Pack is assigned a unique qualification pack code.
Qualifications Pack Code	Qualifications Pack Code is a unique reference code that identifies a qualifications pack.
Scope	Scope is the set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on the quality of performance required.
Sector	Sector is a conglomeration of different business operations having similar businesses and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.



Sub-Sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Sub-functions	Sub-functions are sub-activities essential to fulfil the achieving the objectives of the function.
Technical Knowledge	Technical Knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Unit Code	Unit Code is a unique identifier for a NOS unit, which can be denoted with an 'N'
Unit Title	Unit Title gives a clear overall statement about what the incumbent should be able to do.
Vertical	Vertical may exist within a sub-sector representing different domain areas or the client industries served by the industry.
Keywords /Terms	Description
NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
OEM	Original Equipment Manufacturer
OS	Occupational Standard(s)
QP	Qualifications Pack
5 S	Technique of maintaining orderliness –Japanese terminology
CP	Control Plan
WI	Work Instructions



ISC/N0706: Prepare for operation of control systems



Overview

This unit is about inspecting the incoming coil according to identification, specifications and prepare control system for carrying out the operation

Unit Code	ISC/N0706
Unit Title (Task)	Prepares for operation of control systems
Description	This unit is about inspecting the incoming coil according to specifications and prepare control system for carrying out the operation at different work stations.
Scope	This OS unit/task covers the following: - Inspect the incoming coil according to specifications as laid out in writing or designed for on –line viewing and understand the given job as per schedule /sequence - Load the coil for operation
Performance Criteria (PC) w.r.t. the Scope	
Element	Performance Criteria
Inspect the incoming coil according to specifications and understand the given job	To be competent, the user/individual on the job must be able to PC1. Read and understand the specifications of the job to be carried out PC2. Inspect the incoming coil for dimensions, telescopicity, quality/ physical defect according to the process chart and ensure they tally with PDI (Primary Data Input) PC3. Take necessary actions to get any deviations from specifications corrected PC4. Report any issues that may be out of his/her scope to supervisor
Load the coil for operation	To be competent, the user/individual on the job must be able to: PC5. Understand how to load the coil on the rolling system correctly PC6. Load or stack the coil(s) for conducting the operation, as appropriate PC7. Arrange the coil as per diameter and width to ensure correct placement PC8: Maintain the continuity of operation by welding the forward coil with the trailing coil, where appropriate
Element	Knowledge and Understanding
A. Organisational Context (Knowledge of the Company/ Organisation and its processes)	The user/individual on the job needs to know and understand: KA1. Job specific documents e.g. daily work checklist and importance of the same KA2. Safety policy of the company, as applicable to his area of working KA3. Emergency plan of the specific work site KA4. Risk and impact of not following defined Safe Working Practice/work instructions, or SPI (Standard Practice Instructions) KA5. Escalation matrix for reporting identified problems
B. Technical Knowledge	The user/individual on the job needs to know and understand: KB1. Understand the physical properties and the features of coils, the effect of temperature, time, line-speed, burner flames, molten zinc, acid concentration, bath

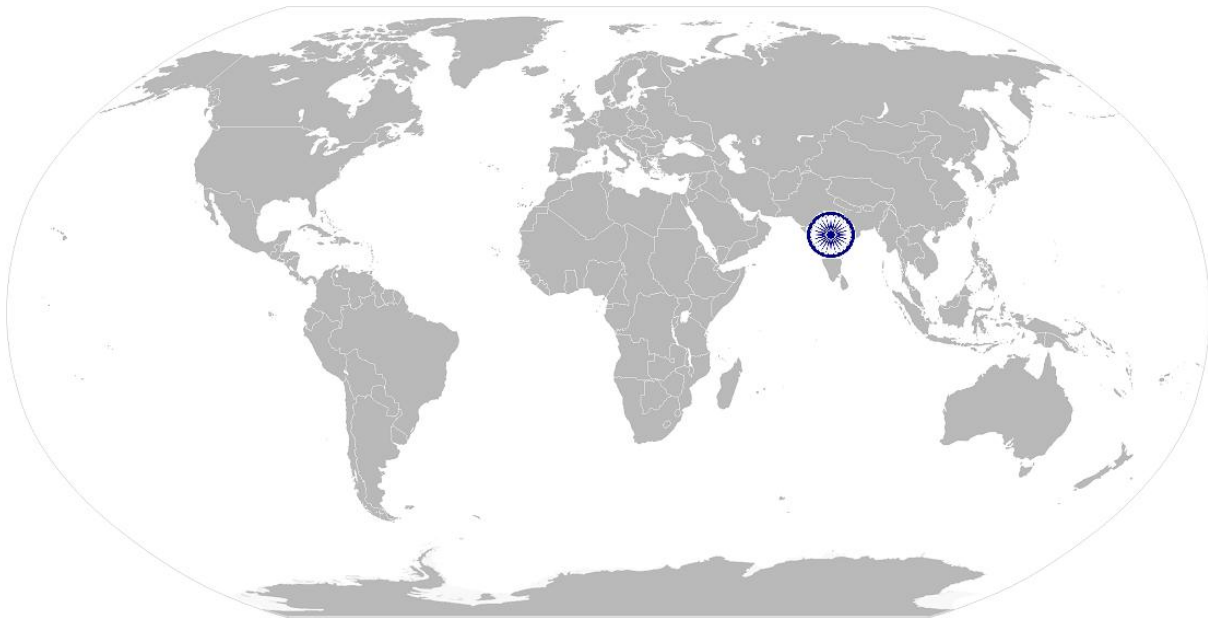


	characteristics, alkali solution etc. on the coils as applicable in the different processing lines. KB2. Read and interpret process charts, control plans, job manuals, Safety and other instruction documents etc. KB3. Understand how the pickling/ rolling/annealing/ electrolytic cleaning/galvanising systems work and the essential associated parts/machines for the specific process KB4. Use of different interlocks and operation sequence
Skills (S) w.r.t. the scope	
Element	Skills
A. Core Skills/ Generic Skills	Writing skills and reading Skills
	The user/individual on the job needs to know and understand how to:
	SA1. Read equipment manuals and process documents given by the equipment supplier to understand the equipment and processes better
	SA2. Make notes on any deviation from desirable specifications/characteristics of the interim processes/work stations and inform supervisor, as required
	Oral Communication (Listening and Speaking skills)
B. Professional Skills	The user/individual on the job needs to know and understand how to:
	SA3. Answer the queries raised by the operative team as well as intra-company departments
	SA4. Discuss task lists, schedules, and work-loads with team members
	Time Management
	The user/individual on the job needs to know and understand how to:
B. Professional Skills	SB1. Prioritize and execute tasks within scheduled time limits
	Reliability
	The user/individual on the job needs to know and understand how to:
	SB2. Avoid absenteeism and maintain quality of work
	SB3. Work in a disciplined environment and adhere to working norms of the organisation
B. Professional Skills	SB4. Be punctual and adhere to timelines



NOS Version Control

NOS Code	ISC/N0706		
Credits (NSQF)	TBD	Version number	1.0
Industry	Iron and steel	Drafted on	18/09/2014
Industry Sub-sector	Steel, Re-Rollers	Last reviewed on	30/12/2014
Occupation	Cold Rolling Mill	Next review date	30/12/2015





ISC/N0707: Perform operation of control systems



Overview

This unit is about operating the control system to give appropriate commands to carry out the operation, checking that the strip matches with the required specifications and then prepare the strip for the next process



Unit Code	ISC/N0707
Unit Title (Task)	Perform operation of control systems
Description	This unit is about operating the control system to give appropriate commands to carry out the operation, checking that the strip matches with the required specifications and then prepare the strip for the next process
Scope	This OS unit/task covers the following: - Check that the strip matches planned/designed specifications - Give appropriate commands for operation - Prepare and dispatch for next operation
Performance Criteria (PC) w.r.t. the Scope	
Element	Performance Criteria
Check the strip matches specifications	To be competent, the user/individual on the job must be able to: PC1. Check and compare the specifications of the coil with those in system/record and ensure correctness of the requirement
Give appropriate commands for operation	To be competent, the user/individual on the job must be able to: PC2. Set appropriate parameters (like pressure, current, speed, time-temperature cycle, fillage, concentration, tension etc.) as applicable for welding / pickling/ rolling/ annealing/ electrolytic cleaning/galvanising according to coil specifications and give command for carrying out the operation PC3. Monitor the process parameters during operation e.g. RPM, temperature, line-tension, pressure, concentration, line-speed, fillage, coating thickness, etc. PC4. Give necessary commands for the next operation (coil cooling, skin-passing, drying, side trimming, chromating, logo printing, oiling etc. as appropriate)
Prepare and dispatch for next operation	To be competent, the user/individual on the job must be able to: PC5. Unload the coil, recoil, weigh, strap, label and dispatch to the next station PC6. Report any issues to the supervisor that may be beyond his/her scope
Element	
A. Organisational Context (Knowledge of the Company/ Organisation and its processes)	The user/individual on the job needs to know and understand: KA1. Job specific documents e.g. specification documents of parameters and importance of the same KA2. Risk and impact of not following defined procedures/work instructions KA3. Escalation matrix for reporting identified problems KA4. Implications of delays in process to the company KA5: Company Safety Rules, as applicable to his area

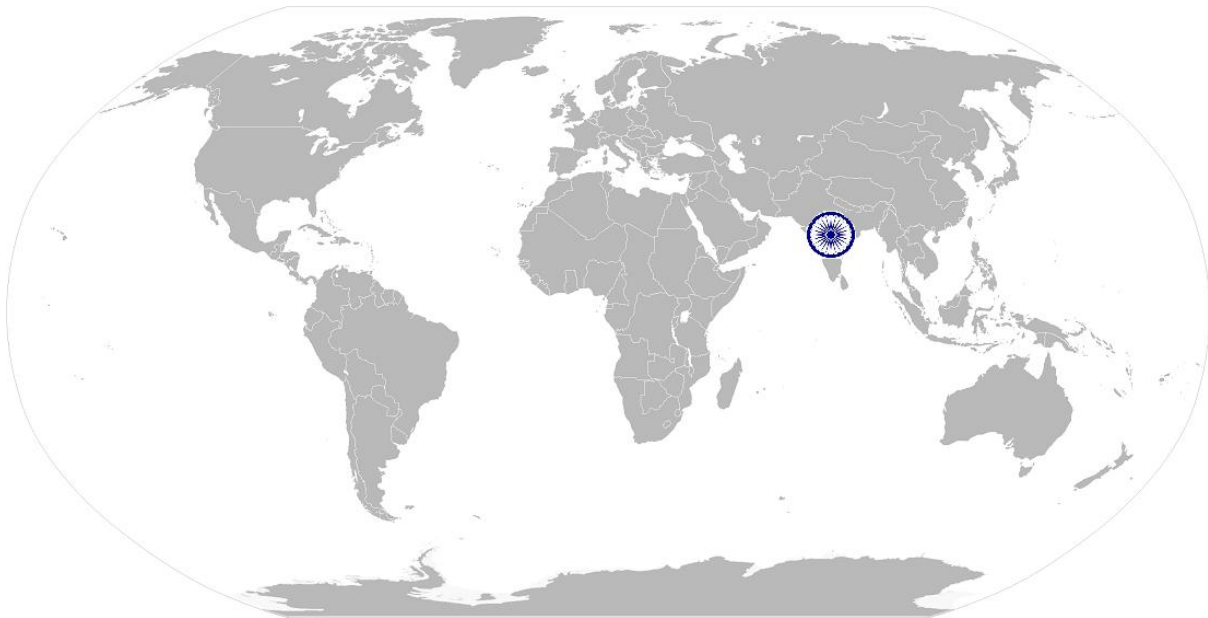


B. Technical Knowledge	The user/individual on the job needs to know/ have the ability to: KB1. Read control parameters on computer screen & take appropriate operating decision accordingly KB2. Operate desktop computer & be conversant with the latest software in use KB3. Knowledge of rolling, pickling, annealing, galvanizing etc. processes along with that of basic metallurgy to understand deeper issues regarding material behaviour KB4. Working knowledge of the Entry, Process and Exit parameters of the continuous process & be familiar with the relevant parameters of any Batch process involved KB5. Identify different parameters & defects to be segregated and noted before passing material e.g. under/over pickling, telescopicity, stickers, dross pick up, surface defects etc.
Skills (S) w.r.t. the scope	
Element	Skills
A. Core Skills/ Generic Skills	Writing skills and reading Skills
	The user/ individual on the job needs to know and understand how to: SA1. Construct simple sentences & write technical observations, memos & requisitions in English, Hindi and Local Language (E/H/LL) SA2. Read equipment manuals and process documents given by the equipment supplier to understand the equipment and processes better SA3. Read, understand, follow/apply memo, reports, instruction manuals, quality control charts and safety documents
	Oral Communication (Listening and Speaking skills)
	The user/individual on the job needs to know and understand how to: SA4. Communicate with supervisors/juniors/colleagues/casual workers in English/Hindi/Local language (E/H/LL) SA5. Respond objectively & orally to questions asked of him in E/H/LL SA6. Discuss task lists, schedules, and work-loads with the team members
B. Professional Skills	Time Management
	The user/individual on the job needs to know and understand how to: SB1. Prioritize and execute tasks within the scheduled time limits
	Reliability
	The user/individual on the job needs to know and understand how to: SB2. Avoid absenteeism and maintain quality of work SB3. Work in a disciplined environment and adhere to working norms of the organisation SB4. Be punctual and adhere to timelines



NOS Version Control

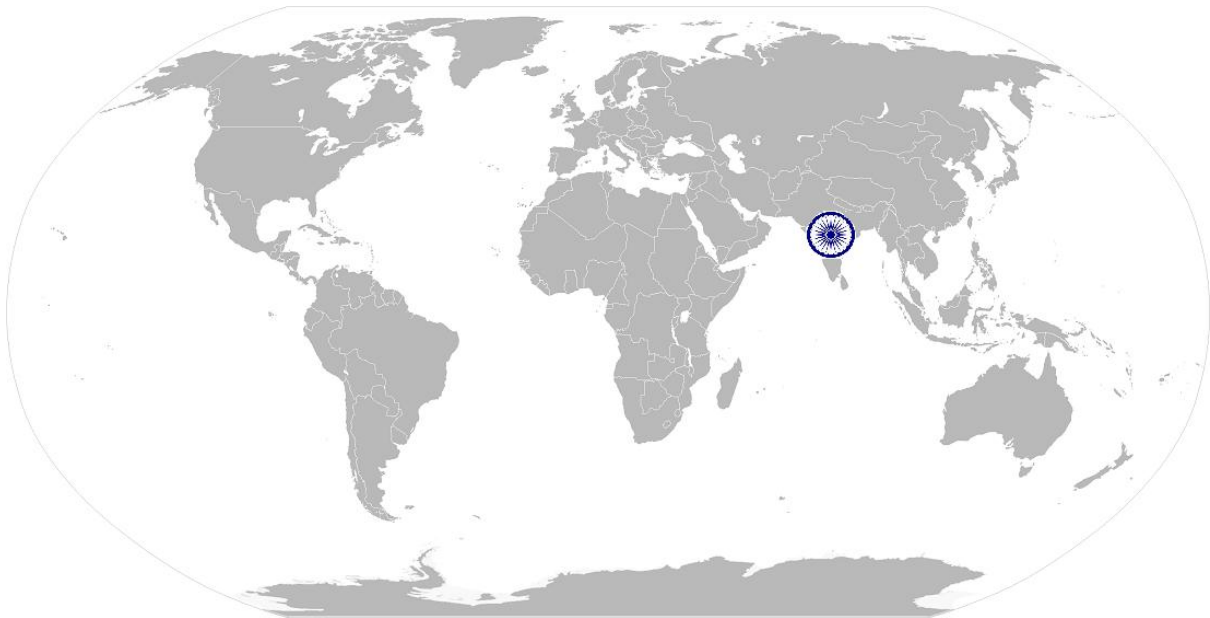
NOS Code	ISC/N0707		
Credits(NSQF)	TBD	Version number	1.0
Industry	Iron and steel	Drafted on	18/09/2014
Industry Sub-sector	Steel, Re-Rollers	Last reviewed on	30/12/2014
Occupation	Cold Rolling Mill	Next review date	30/12/2015





ISC/N0008: Use basic health and safety practices at the workplace

National Occupational Standards



Overview

This unit covers health, safety and security at the workplace. This includes procedures and practices that candidates need to follow to help maintain a healthy, safe and secure work environment.



Unit Code	ISC/N0008
Unit Title (Task)	Use basic health and safety practices at the workplace
Description	This OS unit is about knowledge and practices relating to health, safety and security that candidates need to use in the workplace. It covers responsibilities towards self, others, assets and the environment. It includes understanding of risks and hazards in the workplace, along with common techniques to minimize risk, deal with accidents, emergencies, etc.
Scope	This unit/task covers the following: • Health and safety procedures • Fire safety procedures • Emergencies, rescue and first aid procedures
Performance Criteria (PC) w.r.t. the Scope	
Element	Performance Criteria
Health and safety procedures	The user/individual on the job should be able to: PC1. Use protective clothing/equipment for specific tasks and work conditions Protective clothing includes: • Leather or asbestos gloves • Flame proof aprons • Flame proof overalls buttoned to neck • Cuff less (without folds) trousers • Reinforced footwear • Helmets/hard hats • Cap and shoulder covers • Ear defenders/plugs • Safety boots • Knee pads • Particle masks • Glasses/gloves/visors Equipment includes: • Hand shields • Machine guards • Residual current devices • Shields • Dust sheets • Respirator PC2. State the name and location of people responsible for health and safety in the workplace

Various areas are listed below:

- On chemical containers
- Equipment
- Packages
- Inside buildings
- Open areas, public places etc.

PC3. State the names and location of documents that refer to health and safety in the workplace

PC4. Identify job-site hazardous work and state possible causes of risk or accident in the workplace

Hazards include:

- Working with electrical and thermal tools and equipment
- Sharp edged and heavy tools
- Heated metals
- Oxyfuel and gas cylinders
- Welding radiation
- Surfaces: sharp, slippery, uneven, chipped, broken, etc.
- Substances: chemicals, gas, oxy-fuel, fumes, dust, etc.
- Physical: working at heights, large and heavy objects and machines, sharp and piercing objects, tools and machines, intense light, load noise, obstructions in corridors, by doors, blind turns, noise, over stacked shelves and packages, etc.
- Electrical: power supply and points, loose and naked cables and wires, electrical machines and appliances, etc.

PC5. Carry out safe working practices while dealing with hazards to ensure the safety of self and others state methods of accident prevention in the work environment of the job role

Safe working practices include:

- Using protective clothing and equipment
- Putting up and reading safety signs
- Handle tools in the correct manner and store and maintain them properly
- Keep work area clear of clutter, spillage and unsafe object lying casually
- While working with electricity take all electrical precautions like insulated clothing, adequate equipment insulation, use of control equipment, dry work area, switch off the power supply when not required, etc.
- Safe lifting and carrying practices
- Use equipment that is working properly and is well maintained
- Take due measures for safety while working in confined places, trenches or at heights, etc. Including safety harness, fall arrestors etc.

Methods are:

- Training in health and safety procedures
- Using health and safety procedures
- Use of equipment and working practices (such as safe carrying procedures)
- Safety notices, advice
- Instruction from colleagues and supervisors

	PC6. State location of general health and safety equipment in the workplace PC7. Inspect for faults, set up and safely use steps and ladders in general use Faults : • Corrosion of metal components • Deterioration • Splits and cracks timber components • Imbalance • Loose rungs • Nuts or bolts, etc. Set up: • Firm/level base • Clip/lash down • Leaning at the correct angle, etc. PC8. Work safely in and around trenches, elevated places and confined areas PC9. Lift heavy objects safely using correct procedures PC10. Apply good housekeeping practices at all times. Good housekeeping practices: • Clean/tidy work areas • Removal/disposal of waste products • Protect surfaces PC11. Identify common hazard signs displayed in various areas PC12. Retrieve and/or point out documents that refer to health and safety in the workplace
Fire safety procedures	The user/individual on the job should be able to: PC13. Use the various appropriate fire extinguishers on different types of fires correctly. Fire extinguishers: • Sand • Water • Foam • Co2 • Dry powder Fires: • Class A: Ordinary solid combustibles, e.g. wood, paper, cloth, plastic, charcoal etc. • Class B: Flammable liquids and gases, e.g. gasoline, propane, diesel fuel, tar, cooking oil and similar substances • Class C: Electrical equipment e.g. appliances, wiring, breaker panels etc. (these categories of fires become Class A, B, and D fires when the electrical equipment that initiated the fire is no longer receiving electricity) • Class D: Combustible metals such as magnesium, titanium, and sodium (these fires burn at extremely high temperatures and require special suppression agents) Causes of fires: • Heating of metal

	• Spontaneous ignition • Sparking, • Electrical heating • Loose fires (e.g. Smoking, welding, etc.) • Chemical fires, etc. PC14. Demonstrate rescue techniques applied during fire hazard PC15. Demonstrate good housekeeping in order to prevent fire hazards PC16. Demonstrate the correct use of a fire extinguisher
Emergencies, rescue and first-aid procedures	The user/individual on the job should be able to: PC17. Demonstrate how to free a person from electrocution PC18. Administer appropriate first aid to victims as required e.g. in case of bleeding, burns, choking, electric shock, poisoning etc. PC19. Demonstrate basic techniques of bandaging PC20. Respond promptly and appropriately to an accident situation or medical emergency in real or simulated environments. Few general health and safety equipment are mentioned below : • Fire extinguishers • First aid equipment • Safety instruments and clothing • Safety installations, e.g. Fire exits, exhaust fans etc. PC21. Perform and organize loss minimization or rescue activity during an accident in real or simulated environments PC22. Administer first aid to victims in case of a heart attack or cardiac arrest due to electric shock, before the arrival of emergency services in real or simulated cases PC23. Demonstrate the artificial respiration and the CPR Process PC24. Participate in emergency procedures. Emergency procedures are: • Raising alarm • Safe/efficient evacuation • Correct means of escape • Correct assembly point • Roll call • Correct return to work PC25. Complete a written accident/incident report or dictate a report to another person, and send report to person responsible Incident Report should capture: • Name • Date/time of incident • Date/time of report, • Location • Environment conditions • Persons involved • Sequence of events • Injuries sustained • Damage sustained • Actions taken • Witnesses

	- Supervisor/manager notified Documents: - Fire notices - Accident reports - Safety instructions for equipment and procedures - Company notices and documents - Legal documents (e.g. Government notices) Job titles: - Health and safety officer - First aid officer - Fire officer PC26. Demonstrate correct method to move injured people and others during an emergency
Element	Knowledge and Understanding
A. Organisational Context (Knowledge of the Company/ Organisation and its processes)	The user/individual on the job needs to know and understand: KA1. State the names (and job titles if applicable), and describe where to find, all the people responsible for health and safety in a workplace KA2. State the names and location of documents that refer to health and safety in the workplace
B. Technical Knowledge	The user/individual on the job needs to know and understand: KB3. Meaning of “hazards” and “risks” KB4. Health and safety hazards commonly present in the work environment and related precautions KB5. Possible causes of risk, hazard or accident in the workplace and why risk and/or accidents are possible KB6. Activities and causes of risk and accident KB7. Methods of accident prevention KB8. Safe working practices when working with tools and machines KB9. Safe working practices while working at various hazardous sites KB10. Where to find all the general health and safety equipment in the workplace KB11. Various dangers associated with the use of electrical equipment KB12. Preventative and remedial actions to be taken in the case of exposure to toxic materials - Exposure: ingested, contact with skin, inhaled - Preventative action: ventilation, masks, protective clothing/equipment - Remedial action: immediate first aid, report to supervisor - Materials: solvents, flux, lead KB13. Importance of using protective clothing/equipment while working KB14. Precautionary activities to prevent the fire accident Activities and causes: - Physical actions - Reading - Listening to and giving instructions

	• Inattention • Sickness and incapacity (e.g. Drunkenness) • Health hazards (e.g. Untreated injuries and contagious illness) KB15. Various causes of fire KB16. Techniques of using the different fire extinguishers KB17. Different methods of extinguishing fire KB18. Rescue techniques applied during a fire hazard KB19. Various types of safety signs and what they mean KB20. Appropriate basic first aid treatment relevant to the condition e.g. Shock, electrical shock, bleeding, breaks to bones, minor burns, resuscitation, poisoning, eye injuries KB21. Content of written accident report KB22. Potential injuries and ill health associated with incorrect manual handling KB23. Safe lifting and carrying practices KB24. Personal safety, health and dignity issues relating to the movement of a person by others KB25. Potential impact to a person who is moved incorrectly
Skills (S) w.r.t. the scope	
Element	Skills
A. Core Skills/ Generic Skills	Reading and Writing Skills
	The user/individual on the job needs to know and understand how to:
	SA1. Read and comprehend basic content to read labels, charts, signage's SA2. Read and comprehend basic English to read manuals of operations SA3. Read and write an accident/incident report in local language or English
	Oral Communication (Listening and Speaking skills)
	The user/individual on the job needs to know and understand how to:
B. Professional Skills	SA4. Question co-workers appropriately in order to clarify instructions and other issues SA5. Give clear instructions to co-workers, subordinates others
	Decision Making
	The user/individual on the job needs to know and understand how to:
B. Professional Skills	SA6. Make appropriate decisions pertaining to the concerned area of work with respect to intended work objective, span of authority, responsibility, laid down procedure and guidelines
	Plan and Organize
B. Professional Skills	The user/individual on the job needs to know and understand:
	SB1. Plan and organize their own work schedule, work area, tools, equipment and materials to maintain decorum and for improved productivity

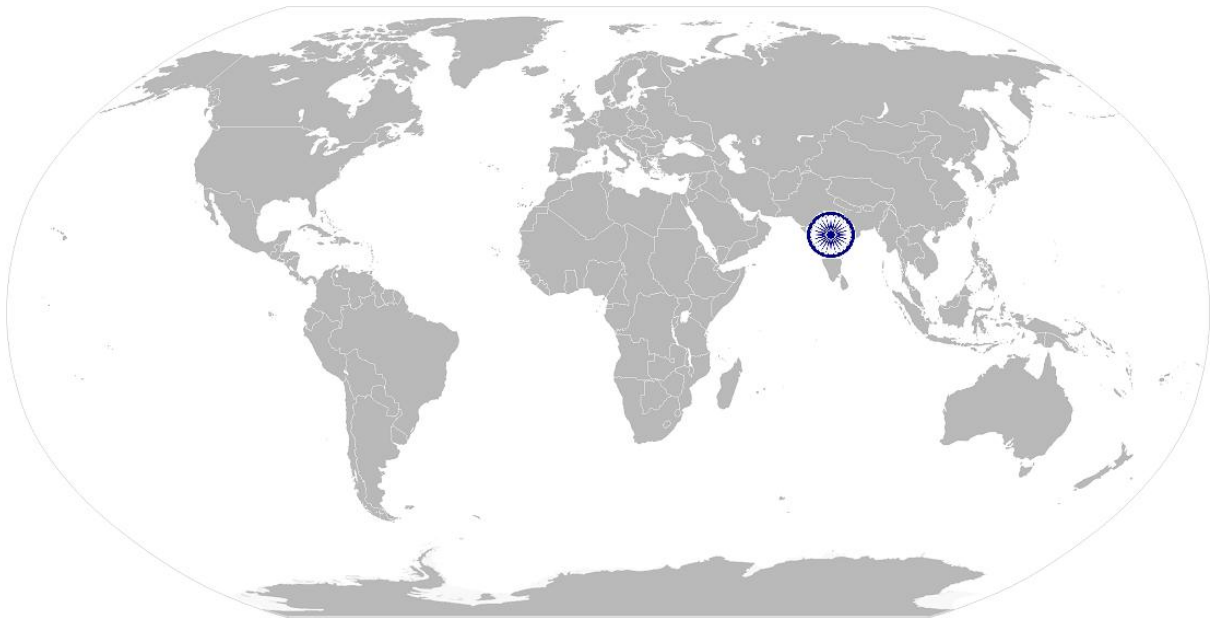


	Working with others
	The user/individual on the job needs to know and understand how to: SB2. Remain congenial while discussing and debating issues with co-workers SB3. Follow appropriate protocols for communication based on situation, hierarchy, organizational culture and practice SB4. Ask for, provide and receive required assistance where possible to ensure achievement of work related objectives SB5. Thank co-workers for any assistance received SB6. Offer appropriate respect based on mutuality and respect for fellow workmanship and authority
	Problem Solving
	The user/individual on the job needs to know and understand how to: SB7. Think through the problem, evaluate the possible solution(s) and suggest an optimum /best possible solution(s) SB8. Identify immediate or temporary solutions to resolve delays SB9. Identify sources of support that can be availed of for problem solving for various kind of problems SB10. Seek appropriate assistance from other sources to resolve problems SB11. Report problems that you cannot resolve to appropriate authority
	Analytical Thinking
	The user/individual on the job needs to know and understand how to: SB12. Identify cause and effect relations in their area of work SB13. Use cause and effect relations to anticipate potential problems and their solution



NOS Version Control

NOS Code	ISC/N0008		
Credits(NSQF)	TBD	Version number	1.0
Industry	Iron and steel	Drafted on	23/07/2014
Industry Sub-sector	All departments	Last reviewed on	30/12/2014
Occupation	Cold Rolling Mill	Next review date	30/12/2015





ISC/N0096: Works effectively with others and manage team members

National Occupational Standards



Overview

This unit covers basic practices that improve effectiveness of working with others in an organisational set-up and managing team members



Unit Code	ISC /N0096
Unit Title (Task)	Works effectively with others and manage team members
Description	This unit covers basic etiquettes and competencies that a candidate is required to possess and demonstrate in their behaviour and interactions with others at the workplace. These cover areas such as communication etiquette, discipline, listening, handling conflict and grievances.
Scope	This unit/task covers the following: - Communicate effectively with team members, co-workers and supervisors - Encourage friendly working environment - Use appropriate titles and terms of respect - Follow code of conduct and ethics as applicable - Follow organisation specific policies - Demonstrate effective behaviours for team work - Ensure engagement of team members through on-the job handholding & support
Performance Criteria (PC) w.r.t. the Scope	
Element	Performance Criteria
Communicate effectively with co-workers and supervisors	The user/individual on the job should be able to: PC1. Accurately receive, absorb and share information and instructions from the supervisor and fellow workers, getting clarification where required PC2. Display appropriate communication etiquette while working PC3. Display active listening skills while interacting with others at work PC4. Use appropriate tone, pitch and language to convey politeness, assertiveness, care and professionalism
Demonstrate effective behaviours for team work	The user/individual on the job should be able to: PC5. Display helpful behaviour by assisting others in performing tasks in effective manner PC6. Consult with and assist others to maximize effectiveness and efficiency in carrying out tasks PC7. Demonstrate responsible and disciplined behaviours at the workplace PC8. Escalate grievances and problems to superiors
Ensure engagement of team members through on-the job handholding & support	The user/individual on the job should be able to: PC9. Communicate day-to-day objectives, instructions etc. to team members PC10. Guide the team members to manage day-to-day issues at work PC11. Gather concerns, feedback from team members and convey them to appropriate authorities



	PC12. Escalate grievances and problems to superiors
Element	Knowledge and Understanding
A. Organisational Context (Knowledge of the Company/ Organisation and its processes)	The user/individual on the job needs to know and understand: KA1. Legislation, standards, policies, and procedures followed in the company relevant to own employment and performance conditions KA2. Reporting structure, inter-dependent functions, lines and procedures in the work area KA3. Relevant people and their responsibilities within the work area KA4. Escalation matrix and procedures for reporting work and employment related issues
B. Technical Knowledge	The user/individual on the job needs to know and understand: KB1. Various categories of people that one is required to communicate and co-ordinate with in the organization KB2. Importance of teamwork in organizational and individual success KB3. Various components of effective communication and importance of effective communication while communicating professional circles KB4. Key elements of active listening and importance of active listening KB5. Importance of ethics for professional success KB6. Importance of discipline and disciplined behaviour for professional success KB7. Common reasons, importance and ways of managing interpersonal conflict effectively KB8. Expressing and addressing grievances appropriately and effectively
Skills (S) w.r.t. the scope	
Element	Skills
A. Core Skills/ Generic Skills	Writing skills and reading skills The user/individual on the job needs to know and understand how to: SA1. Read and comprehend basic content to read instructions, labels, charts, signage's SA2. Write notes in log books etc. SA3. Prepare incident reports, as appropriate
	Oral Communication (Listening and Speaking skills) The user/individual on the job needs to know and understand how to: SA4. Question co-workers appropriately in order to clarify instructions and other issues SA5. Communicate with co-workers SA6. Share information with Supervisors
	Decision Making

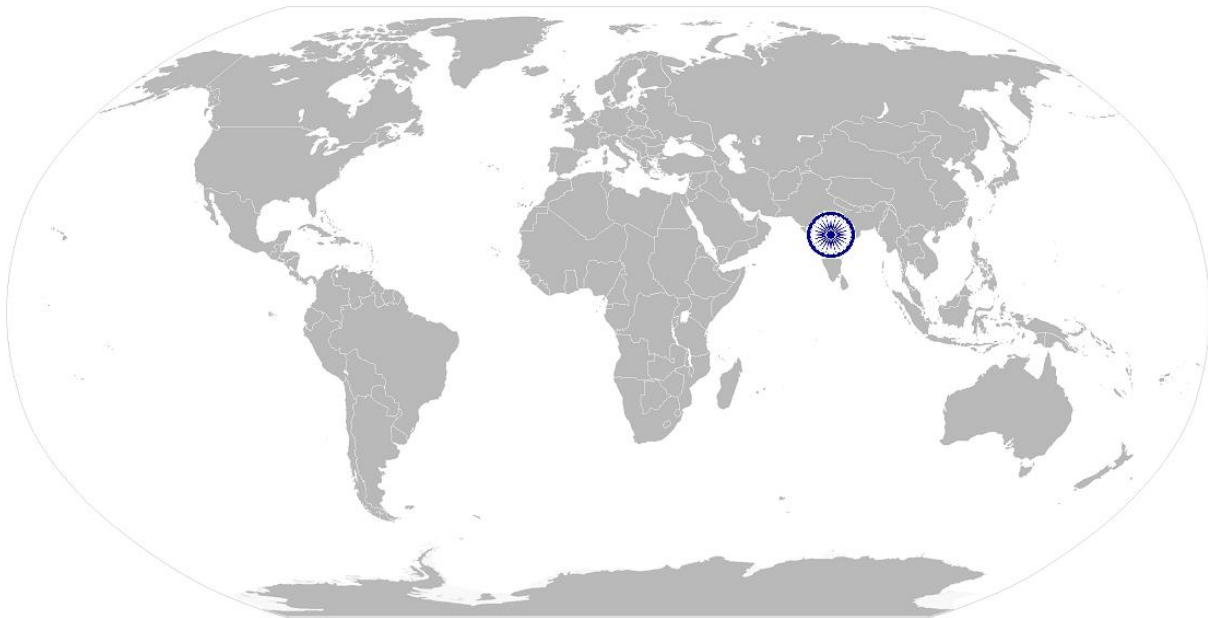


	The user/individual on the job needs to know and understand how to: SA4. Make appropriate decisions pertaining to the concerned area of work with respect to safely achieving work objective following the laid down procedure and guidelines
B. Professional Skills	Plan and Organize
	The user/individual on the job needs to know and understand: SB1. Plan and organize their own work schedule, work area, tools, equipment and materials to maintain decorum and for improved productivity
	Problem Solving
	The user/individual on the job needs to know and understand how to: SB2. Identify sources of support that can be availed of for problem solving for various kind of problems SB3. Seek appropriate assistance from other sources to resolve problems SB4. Report problems that cannot be resolved to appropriate authority
	Analytical Thinking
	The user/individual on the job needs to know and understand how to: SB5. Identify cause and effect relations in own area of work SB6. Use cause and effect relations to anticipate potential problems and their solution



NOS Version Control

NOS Code	ISC/N0096		
Credits(NSQF)	TBD	Version number	1.0
Industry	Iron and steel	Drafted on	23/07/2014
Industry Sub-sector	All departments	Last reviewed on	30/12/2014
Occupation	Cold Rolling Mill	Next review date	30/12/2015





CRITERIA FOR ASSESSMENT OF TRAINEES

Job Role: Process Operator Rolling Mills

Qualification Pack: ISC/Q0701

Sector Skill Council: Indian Iron & Steel Sector Skill Council

Guidelines for Assessment:

1. Criteria for assessment for each Qualification Pack will be created by the Sector Skill Council. Each Performance Criteria (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down proportion of marks for Theory and Skills Practical for each PC.
2. The assessment for the theory part will be based on knowledge bank of questions created by the SSC.
3. Individual assessment agencies will create unique question papers for theory part for each candidate at each examination/training center (as per assessment criteria below)
4. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/training center based on this criteria.
5. To pass the Qualification Pack, every trainee should score a minimum of 60% in every NOS.
6. In case of successfully passing only certain number of NOS's, the trainee is eligible to take subsequent assessment on the balance NOS's to pass the Qualification Pack.

NOSs	PCs	Marks Allocated			
		Total Marks 1000	Out of	Theory	Practical
ISC/N0706: Prepare for operation of control systems	PC1. Read and understand the specifications of the job to be carried out	200	20	5	15
	PC2. Inspect the incoming coil for dimensions, telescopicity, quality/ physical defect according to the process chart and ensure they tally with PDI (Primary Data Input)		30	5	25
	PC3. Take necessary actions to get any deviations from specifications corrected		20	5	15
	PC4. Report any issues that may be out of his/her scope to supervisor		15	0	15
	PC5. Understand how to load the coil on the rolling system correctly		25	5	20
	PC6. Load or stack the coil(s) for conducting the operation, as appropriate		20	0	20



	PC7. Arrange the coil as per diameter and width to ensure correct placement		25	5	20
	PC8: Maintain the continuity of operation by welding the forward coil with the trailing coil, where appropriate		45	5	40
		Total	200	30	170
ISC/N0707: Perform operation of control systems	PC 1. Check and compare the specifications of the coil with those in system/record and ensure correctness of the requirement	550	70	20	50
	PC2. Set appropriate parameters (like pressure, current, speed, time-temperature cycle, fillage, concentration, tension etc.) as applicable for welding / pickling/ rolling/ annealing/ electrolytic cleaning/galvanising according to coil specifications and give command for carrying out the operation		250	50	200
	PC3. Monitor the process parameters during operation e.g. RPM, temperature, line- tension, pressure, concentration, line-speed, fillage, coating thickness, etc.		120	20	100
	PC4. Give necessary commands for the next operation (coil cooling, skin-passing, drying, side trimming, chromating, logo printing, oiling etc. as appropriate)		60	10	50
	PC5. Unload the coil, recoil, weigh, strap, label and dispatch to the next station		30	0	30
	PC6. Report any issues to the supervisor that may be beyond his/her scope		20	5	15
		Total	550	105	445
ISC/N0008: Use basic health and safety practices at	PC1. Use protective clothing/equipment for specific tasks and work conditions	150	9	4	5
	PC2. State the name and location of people responsible for health and safety in the workplace		6	1	5



the workplace	PC3. State the names and location of documents that refer to health and safety in the workplace	2	1	1
	PC4. Identify job-site hazardous work and state possible causes of risk or accident in the workplace	8	4	4
	PC5. Carry out safe working practices while dealing with hazards to ensure the safety of self and others state methods of accident prevention in the work environment of the job role	6	1	5
	PC6. State location of general health and safety equipment in the workplace	6	1	5
	PC7. Inspect for faults, set up and safely use steps and ladders in general use	6	1	5
	PC8. Work safely in and around trenches, elevated places and confined areas	6	1	5
	PC9. Lift heavy objects safely using correct procedures	6	1	5
	PC10. Apply good housekeeping practices at all times	2	1	1
	PC11. Identify common hazard signs displayed in various areas	6	5	1
	PC12. Retrieve and/or point out documents that refer to health and safety in the workplace	5	1	4
	PC13. Use the various appropriate fire extinguishers on different types of fires correctly	9	4	5
	PC14. Demonstrate rescue techniques applied during fire hazard	8	4	4
	PC15. Demonstrate good housekeeping in order to prevent fire hazards	2	1	1
	PC16. Demonstrate the correct use of a fire extinguisher	6	1	5
	PC17. Demonstrate how to free a person from electrocution	6	1	5
	PC18. Administer appropriate first aid to victims as required e.g. in case of bleeding, burns, choking, electric shock, poisoning etc.	8	3	5



	PC19. Demonstrate basic techniques of bandaging		6	1	5
	PC20. Respond promptly and appropriately to an accident situation or medical emergency in real or simulated environments		7	2	5
	PC21. Perform and organize loss minimization or rescue activity during an accident in real or simulated environments		6	1	5
	PC22. Administer first aid to victims in case of a heart attack or cardiac arrest due to electric shock, before the arrival of emergency services in real or simulated cases		6	1	5
	PC23. Demonstrate the artificial respiration and the CPR Process		6	1	5
	PC24. Participate in emergency procedures		6	1	5
	PC25. Complete a written accident/incident report or dictate a report to another person, and send report to person responsible		4	1	3
	PC26. Demonstrate correct method to move injured people and others during an emergency		2	1	1
		Total	150	45	105
ISC/N0096: Works effectively with others and manage team members	PC1. Accurately receive, absorb and share information and instructions from the supervisor and fellow workers, getting clarification where required	100	5	0	5
	PC2. Display appropriate communication etiquette while working		10	0	10
	PC3. Display active listening skills while interacting with others at work		10	0	10
	PC4. Use appropriate tone, pitch and language to convey politeness, assertiveness, care and professionalism		5	0	5
	PC5. Display helpful behaviour by assisting others in performing tasks in effective manner		10	0	10



PC6. Consult with and assist others to maximize effectiveness and efficiency in carrying out tasks	10	0	10
PC7. Demonstrate responsible and disciplined behaviours at the workplace	5	0	5
PC8. Escalate grievances and problems to superiors	5	0	5
PC9. Communicate day-to-day objectives, instructions etc. to team members	15	5	10
PC10. Guide the team members to manage day-to-day issues at work	5	0	5
PC11. Gather concerns, feedback from team members and convey them to appropriate authorities	15	5	10
PC12. Escalate grievances and problems to superiors	5	0	5
Total	100	10	90

